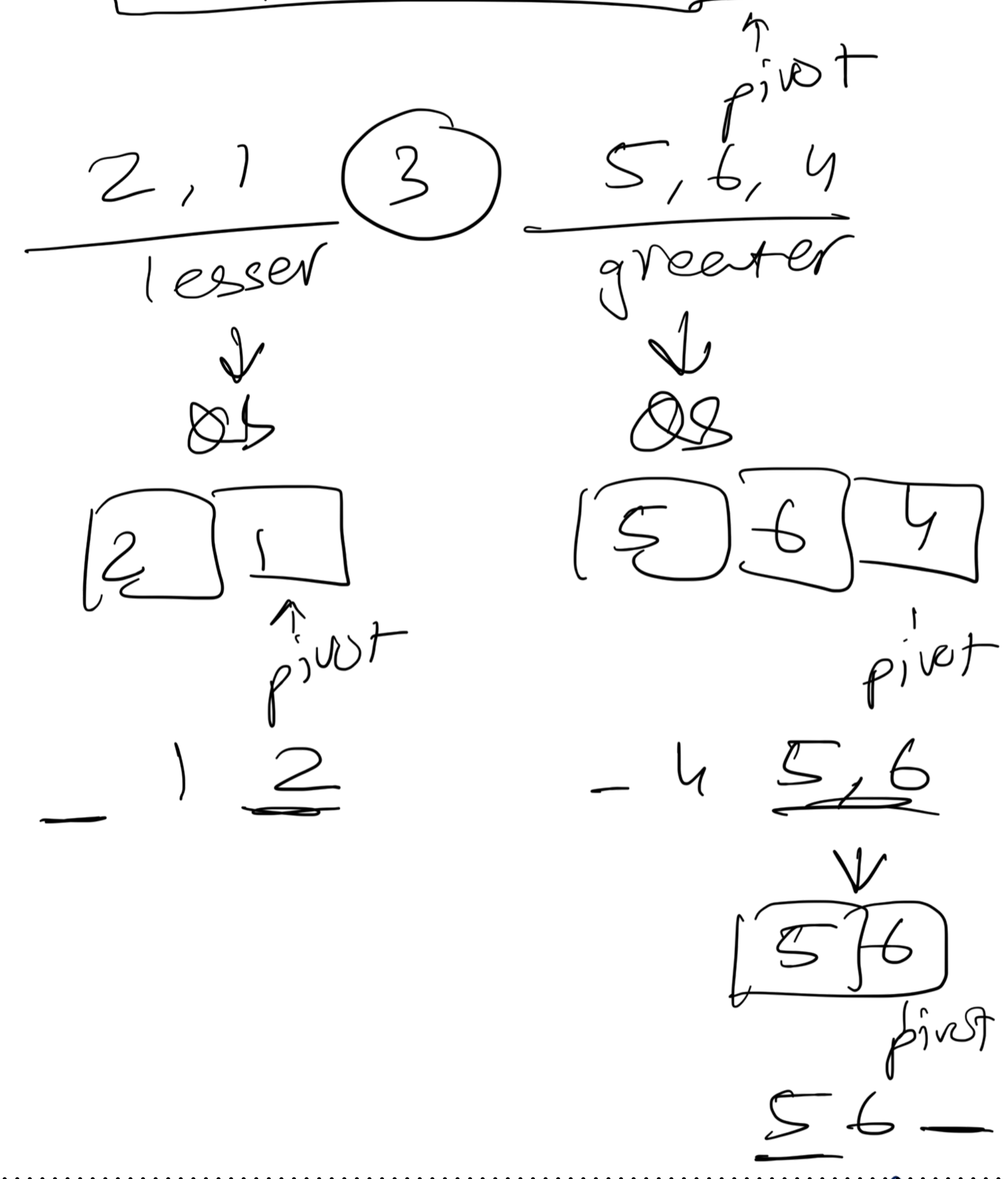
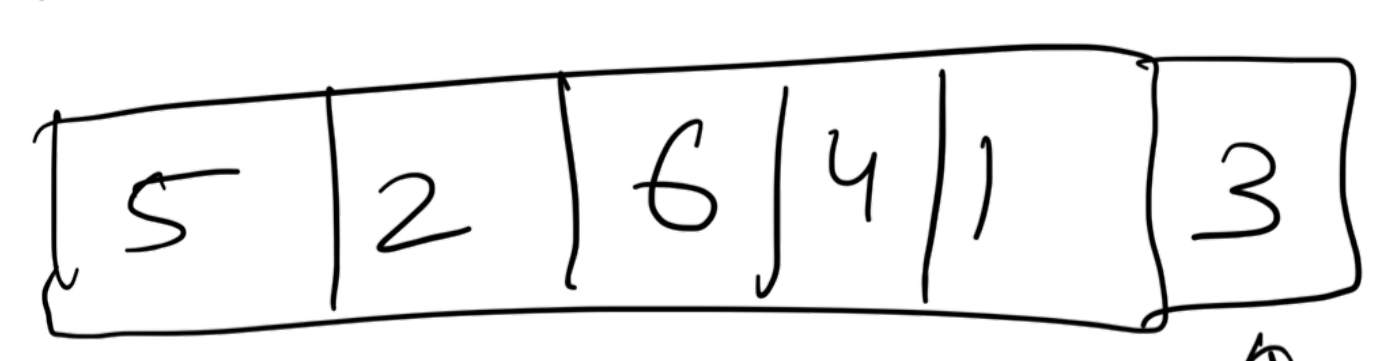
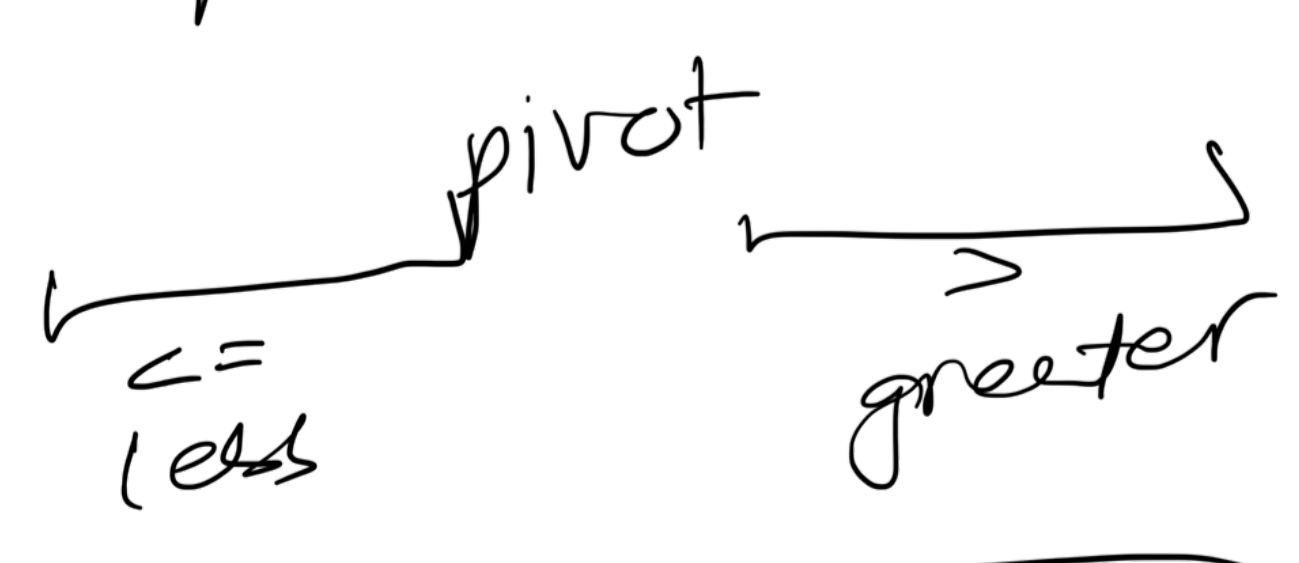


Quick Sort
 pivot & partition
 special idn (first, last...)

- pick pivot
- partition
- QS (left half)
- QS (right half)

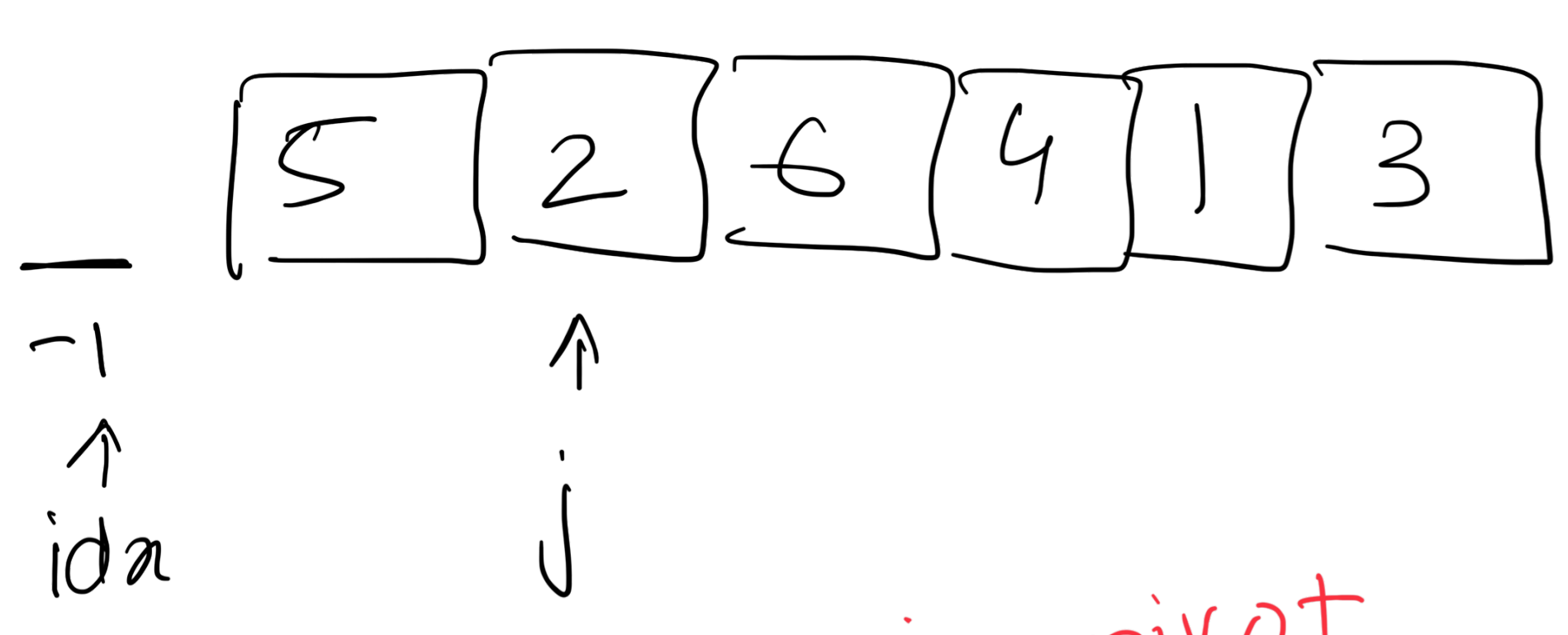
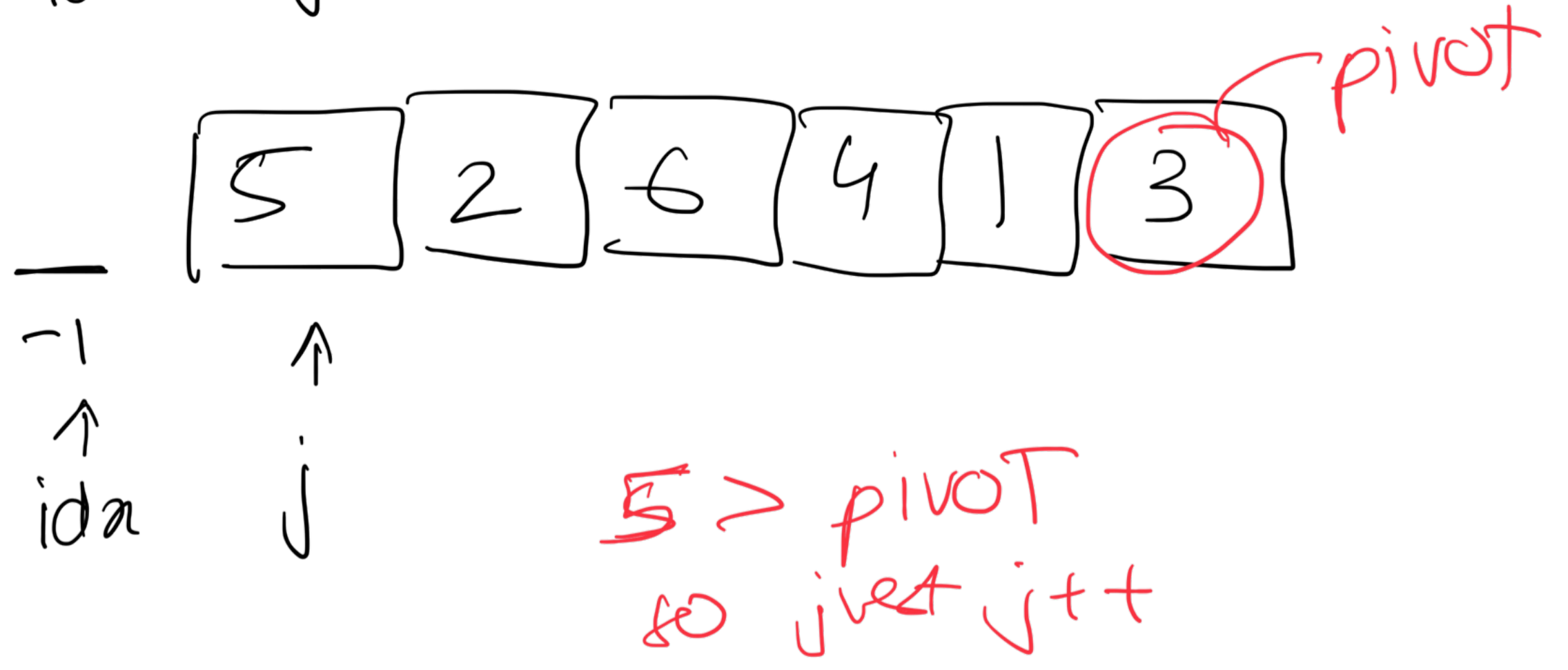
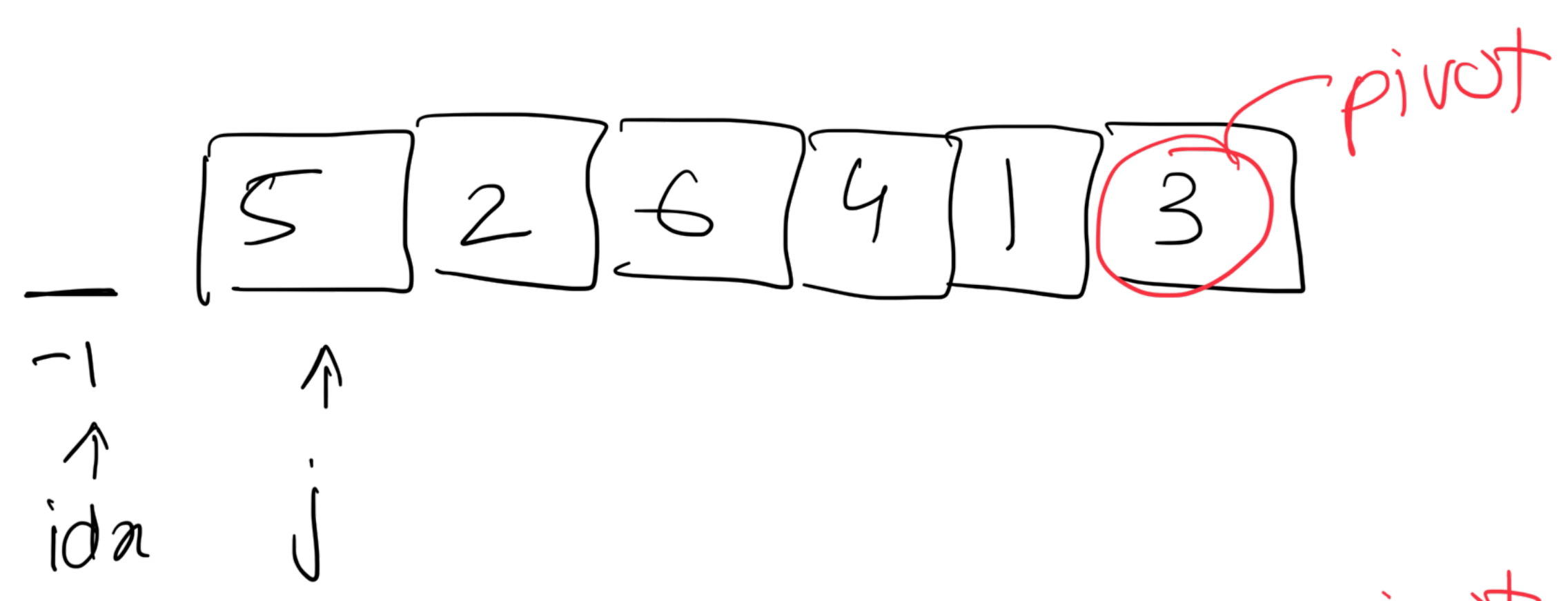


```
void quicksort(arr, st, end) {
  if (st < end) {
    pivIdn = partition(arr, st, end)
    QS(arr, st, pivIdn - 1) //left
    QS(arr, pivIdn + 1, end) //right
  }
}
```

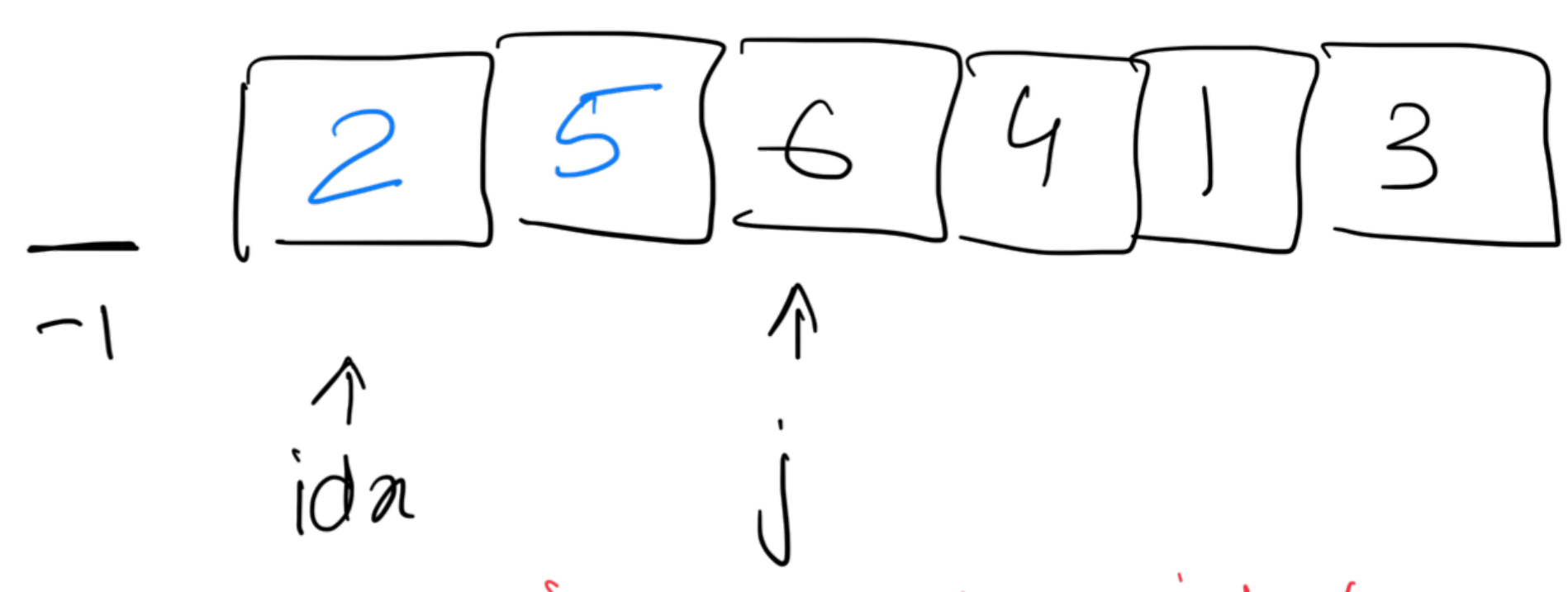
→ returns the true index of the pivot so we can gauge what is left half and right half

Partition

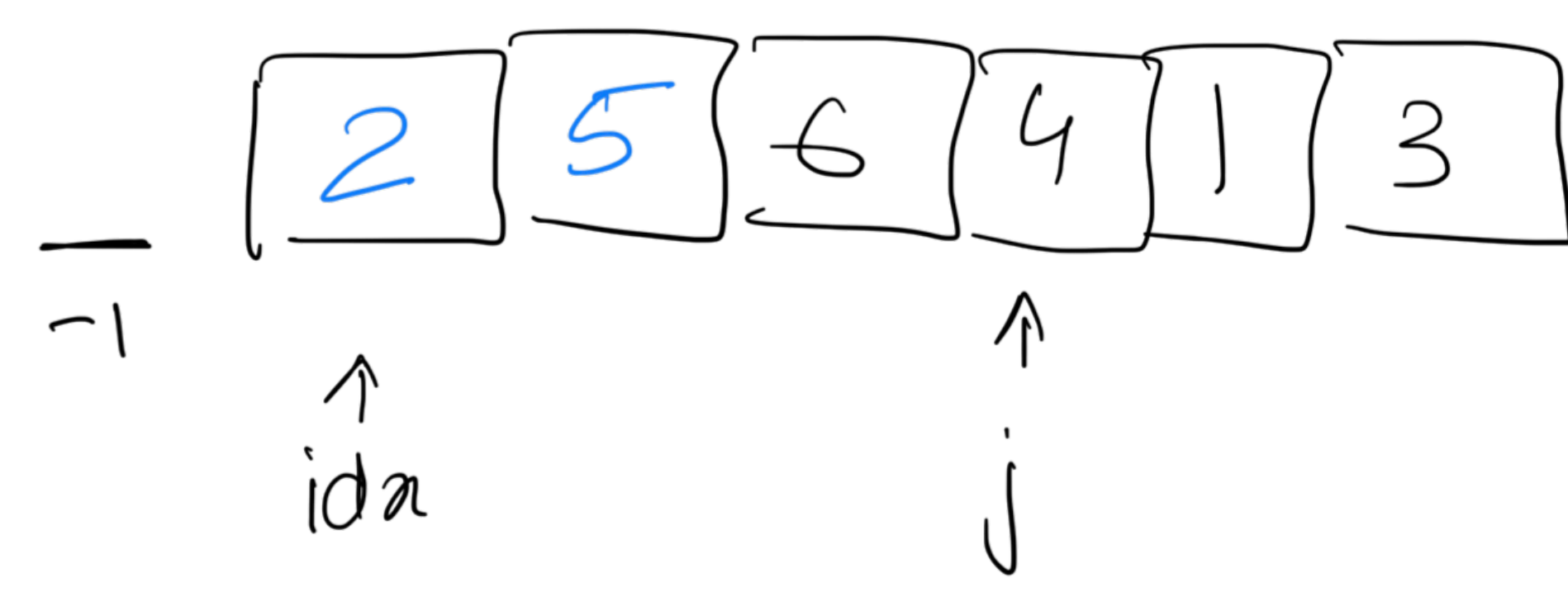
```
void partition(arr, st, end) {
  idn = st - 1, pivot = arr[end]
  for (j = st; j < end; j++) {
    if (arr[j] <= pivot) {
      idn++
      swap(arr[j], arr[idn])
    }
  }
  idn++
  swap(arr[end], arr[idn])
  return idn
}
```



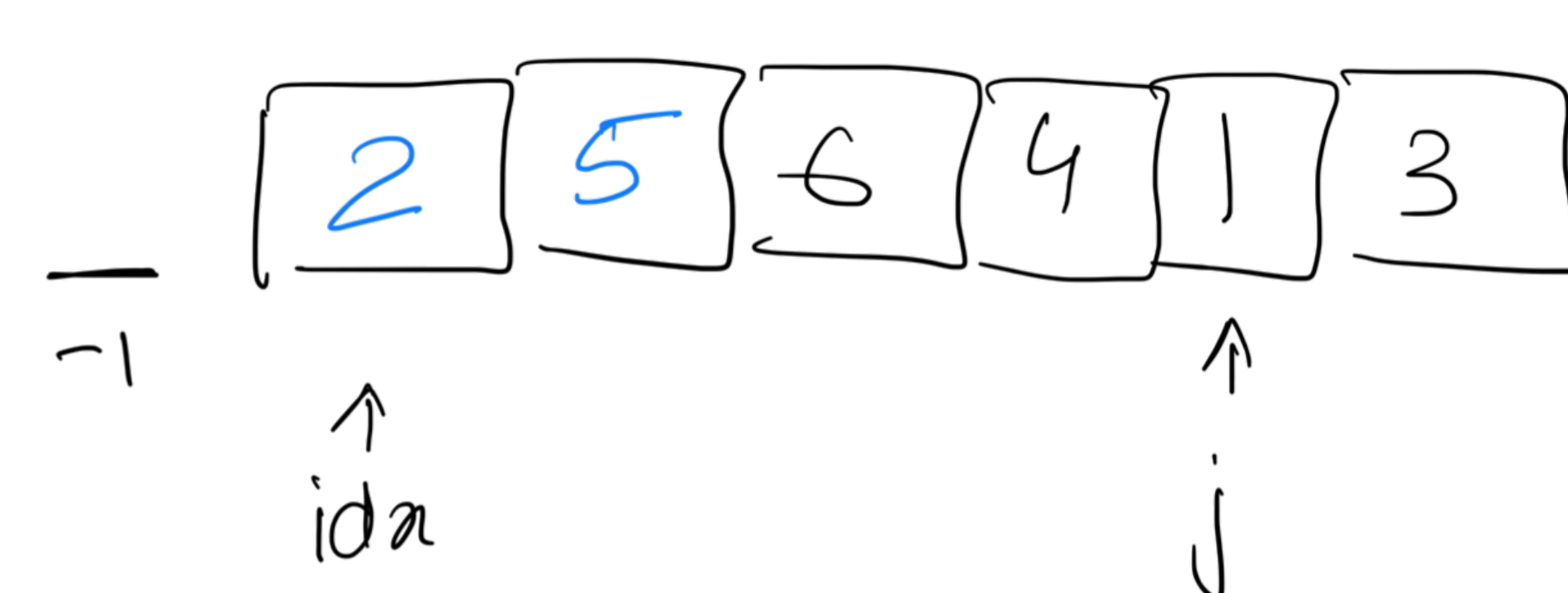
now $j < \text{pivot}$
 $\text{idn}++$
 swap(idn, j)



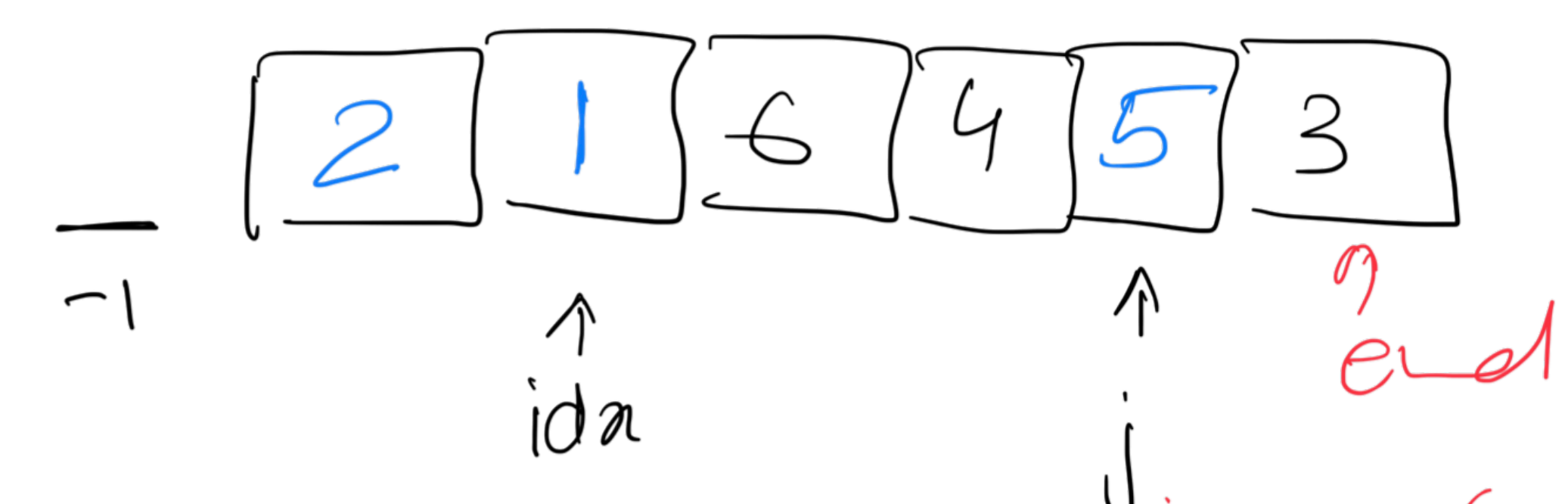
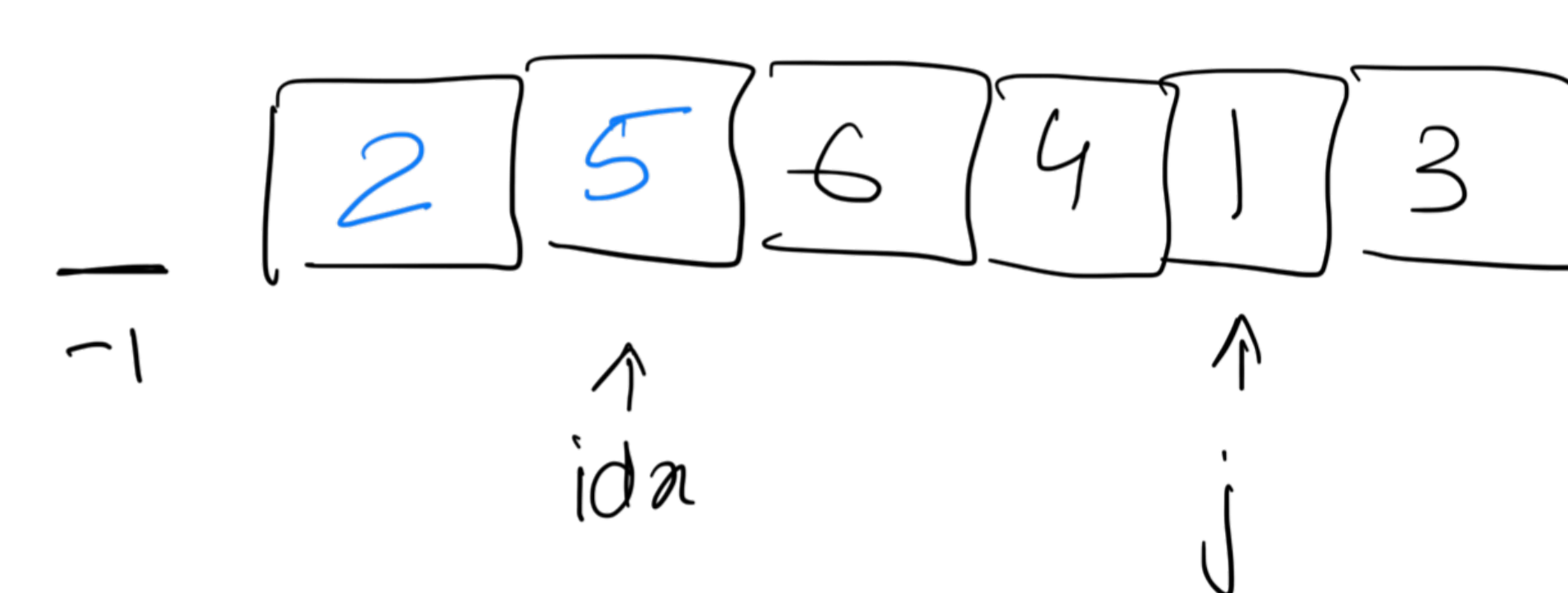
$j > \text{pivot}$ $j++$



$j > \text{pivot}$ $j++$



$j < \text{pivot}$
 $\text{idn}++$
 swap(idn, j)



how get correct position of pivot
 $\text{idn}++$
 swap(end, idn)

